

Sovereign AI Workbench

Prototype Benchmark & Hardware Validation Report

SIH Problem Statement: SIH26117 — Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work

Team: CodeFreaks2.0 Benchmark: 27 September 2026, 22:57 IST

1. Executive Summary

This report uses the corrected benchmark run supplied for the prototype. The test successfully benchmarked three local open-weight models through Ollama on an NVIDIA GeForce RTX 5050 8 GB-class GPU: qwen3:4b, qwen2.5-coder:3b and gemma3:4b. Each model was tested three times using the same enterprise-oriented prompt. The reported latency and generation-speed figures below are calculated directly from the supplied benchmark results.

Scope limitation: This benchmark validates local LLM inference only. It does not establish performance for RAG, OCR, VLM/image processing, sandbox execution, artifact generation or fully air-gapped networking because those workloads were not included in the supplied benchmark.

2. Test Environment

Component	Measured / reported value
CPU	AMD Ryzen 7 5800XT 8 core processor
Logical CPU count	16
GPU	NVIDIA GeForce RTX 5050
GPU VRAM	8151 MB (~8 GB)
System RAM	16 GB
GPU driver	591.86
CUDA version reported	13.1
GPU temperature at snapshot	42 °C
GPU utilization at snapshot	0%
Operating System	Windows-11-10.0.26200-SP0
Python	3.14.3
Ollama	0.34.4

3. Model Configuration

Model	Runs	Purpose in prototype	Average generation speed
qwen3:4b	3	General-purpose local LLM	95.825 tok/s
qwen2.5-coder:3b	3	Local coding / code-generation LLM	125.848 tok/s
gemma3:4b	3	General / multimodal-capable local model	96.619 tok/s

4. Inference Performance

Model	Run	TTFT	Response latency	Tokens	Generation speed
qwen3:4b	1	8.9311 s	10.7346 s	578	95.650 tok/s
qwen3:4b	2	3.9132 s	5.6910 s	538	95.909 tok/s
qwen3:4b	3	3.9112 s	5.6888 s	538	95.915 tok/s
qwen2.5-coder:3b	1	7.7268 s	9.9038 s	273	125.496 tok/s
qwen2.5-coder:3b	2	0.0249 s	2.1048 s	262	126.067 tok/s
qwen2.5-coder:3b	3	0.0333 s	2.1147 s	262	125.980 tok/s
gemma3:4b	1	0.0472 s	1.6952 s	159	96.581 tok/s
gemma3:4b	2	0.0437 s	1.6903 s	159	96.655 tok/s
gemma3:4b	3	0.0486 s	1.6960 s	159	96.622 tok/s

5. Aggregated Benchmark Results

Metric	qwen3:4b	qwen2.5-coder:3b	gemma3:4b
Average time to first token	5.585 s	2.595 s	0.046 s
Average response latency	7.371 s	4.708 s	1.694 s
Average generation speed	95.825 tok/s	125.848 tok/s	96.619 tok/s
Average generated tokens	551.3 tokens	265.7 tokens	159.0 tokens

6. Key Results for SIH Presentation

Metric	Final value to cite
Tested GPU / VRAM	NVIDIA GeForce RTX 5050 / 8151 MB reported (~8 GB)
System RAM	16 GB
Ollama version	0.34.4
Local models benchmarked	qwen3:4b, qwen2.5-coder:3b, gemma3:4b
qwen3:4b generation speed	95.825 tok/s average
qwen2.5-coder:3b generation speed	125.848 tok/s average
gemma3:4b generation speed	96.619 tok/s average
Offline validation	Not performed in this benchmark

7. Reproducibility

The benchmark used the same prompt for all selected models and performed three runs per model. The raw benchmark timestamp is 2026-09-27T22:57:05.840524+05:30. The three selected models are explicitly recorded in the benchmark output.

8. Interpretation Notes

The first qwen3:4b and qwen2.5-coder:3b runs show higher time-to-first-token because the model/runtime may incur startup or loading overhead. For example, qwen3:4b TTFT decreases from 8.9311 s on run 1 to about 3.91 s on runs 2–3, while qwen2.5-coder:3b decreases from 7.7268 s to below 0.04 s. These are best reported as cold/warm benchmark observations rather than as a single universal latency figure.

Source: corrected sih_benchmark_results.json supplied for this report.